

PHASE 2 – REQUIREMENT ANALYSIS PHASE

PROJECT

EduGenie – Google Gemini Powered Learning Assistant

OBJECTIVE

To identify the functional and technical requirements needed to develop EduGenie.

PROBLEM STATEMENT

Students often need simple explanations, quick answers, revision materials, quizzes, and guidance for learning a topic. EduGenie is designed to provide these learning-support features through a lightweight web application.

FUNCTIONAL REQUIREMENTS

1. Ask questions and receive AI-generated answers.
2. Explain complex concepts in simple language.
3. Generate quizzes from a topic or text.
4. Summarize long educational passages.
5. Provide personalized learning recommendations from beginner to advanced level.
6. Display results through a simple web interface.

TECHNICAL REQUIREMENTS

1. Python 3.10+
2. FastAPI
3. HTML and CSS
4. Google Gemini API
5. LaMini-Flan-T5-783M
6. Uvicorn
7. Jinja2

PROCEDURE

1. Define the target users as learners/students.
2. List the major learning-support features.
3. Identify the backend and frontend requirements.
4. Select AI models according to the required tasks.
5. Identify API endpoints for each major function.
6. Finalize the requirements for implementation.